

Persistent Organic Pollutants and Type 2 Diabetes: A Prospective Analysis in the Nurses' Health Study and Meta-analysis

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Background: Prospective data regarding persistent organic pollutants (POPs) and risk of type 2 diabetes (T2D) are limited, and the results for individual POPs are not entirely consistent across studies. Objectives: We prospectively examined plasma POP concentrations in relation to incident T2D and summarized existing evidence in a meta-analysis. Methods: Plasma polychlorinated biphenyls (PCBs), dichlorodiphenyltrichloroethane (DDT), dichlorodiphenyldichloroethylene (DDE), and hexachlorobenzene (HCB) concentrations were measured in 1,095 women who were free of diabetes at blood draw in 1989–1990 and participated in two case–control studies in the Nurses' Health Study. We identified 48 incident T2D cases through 30 June 2008. We conducted a literature search in PubMed and EMBASE through December 2011 to identify prospective studies on POPs in relation to diabetes. We used a fixed-effects model to summarize results. Results: After multivariable adjustment, plasma HCB concentration was positively associated with incident T2D [pooled odds ratio (OR) 3.59 (95% CI: 1.49, 8.64, $p_{trend} = 0.003$) comparing extreme tertiles]. Other POPs were not significantly associated with diabetes. After pooling our results with those of six published prospective studies that included 842 diabetes cases in total, we found that HCB and total PCBs both were associated with diabetes: the pooled ORs were 2.00 (95% CI: 1.13, 3.53; $I^2 = 21.4\%$, $p_{heterogeneity} = 0.28$) and 1.70 (95% CI: 1.28, 2.27; $I^2 = 16.3\%$, $p_{heterogeneity} = 0.30$) for HCB and total PCBs, respectively. Conclusions: These findings support an association between POP exposure and the risk of T2D.